

xv6 Basics

1. How many files does the command stressfs create?

Answer: The command stressfs creates 5 files. (Specifically, it spawns 4 children, resulting in 5 processes in total, each of which creates a file named stressfs0, stressfs1, stressfs2, stressfs3, and stressfs4 respectively).

2. What does xv6 print if you run the command wc on itself? That is, the command wc wc.

Answer: Running the command "wc wc" in xv6 prints the following output:

```
73 527 16412 wc
```

3. Where are system call numbers defined and how many are there currently?

Answer: The system call numbers are defined in the file "syscall.h". There are currently 21 system calls defined.

4. In the implementation of the fork system call, what is the local variable name for the pointer to the process control block of the child processes? What about the PCB for the parent process?

Answer: In the implementation of the fork system call (located in "proc.c"):

- The local variable name for the pointer to the child process's PCB is "np".
- The local variable name for the pointer to the parent process's PCB is "curproc".